ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems.

Ex.No.3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation Loss.

Objective :

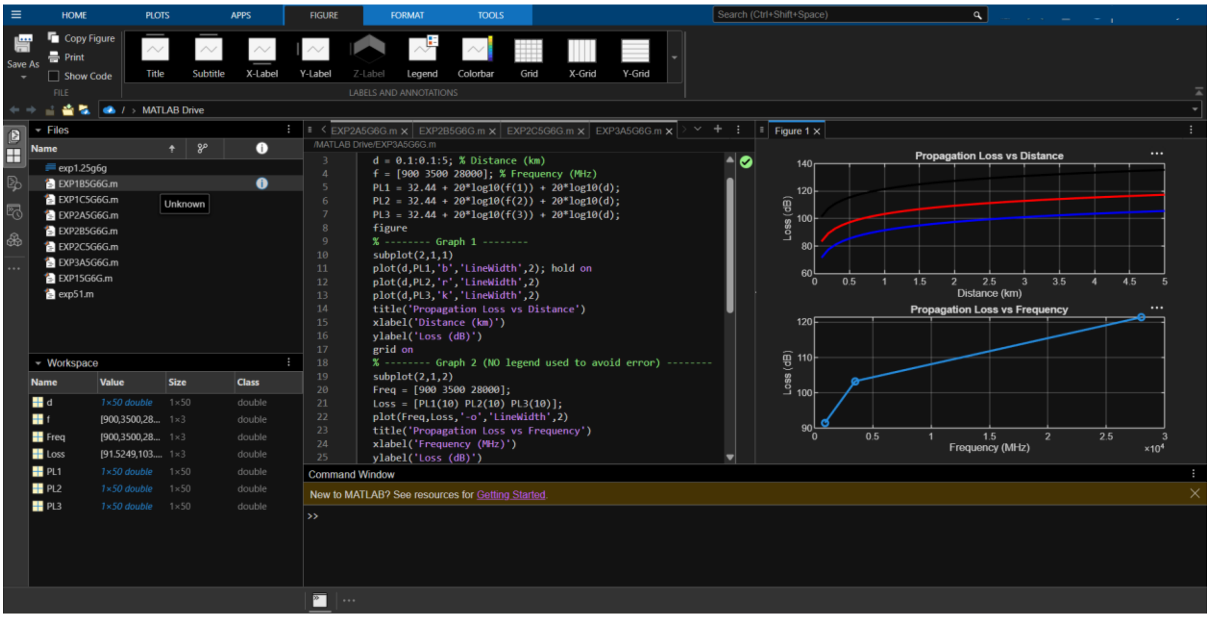
1. To compute the Propagation Loss (dB) for Sub-1 GHz, Mid-band, and mmWave frequency ranges using standard path loss models.
2. To evaluate the impact of different Operating Frequencies (GHz) on signal attenuation and coverage performance.
3. To analyze how Communication Distance (km) affects propagation loss across different spectrum bands.

PROGRAM:

```
clc; clear; close all;
d = 0.1:0.1:5; % Distance (km)
f = [900 3500 28000]; % Frequency (MHz)
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
grid on
```

```
% -------- Graph 2 (NO legend used to avoid error) --------
subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq,Loss,'-o','LineWidth',2)
title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
grid on
```

OUTPUT:



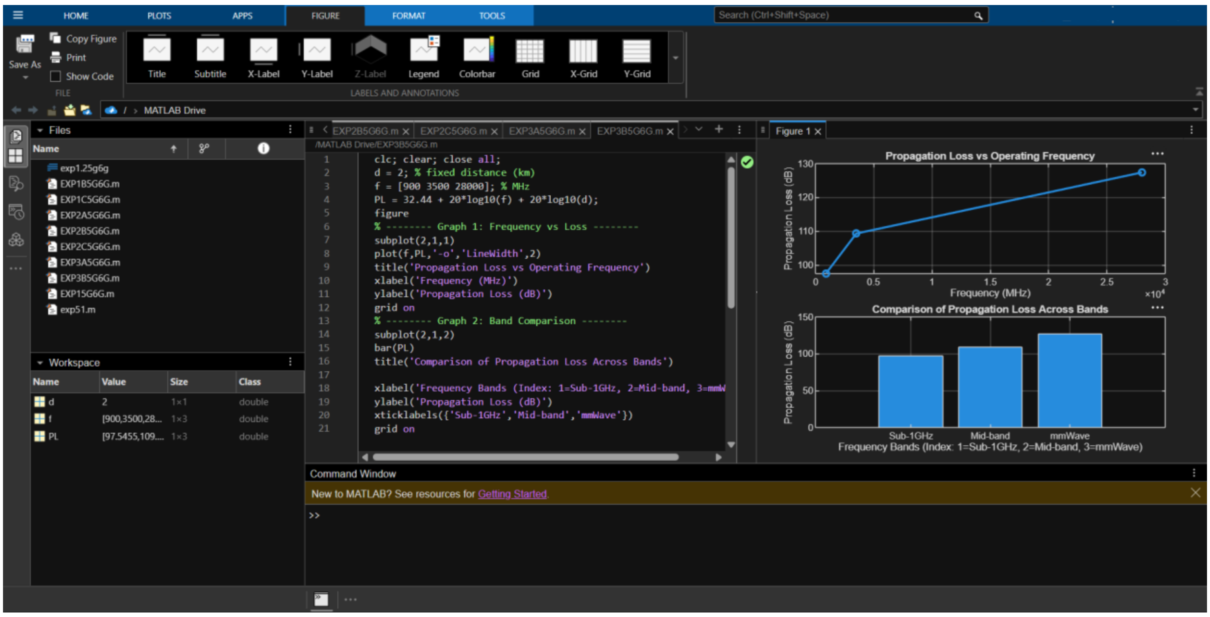
Objective 2: Matlab Code and Output

PROGRAM:

```
clc; clear; close all;
d = 2; % fixed distance (km)
f = [900 3500 28000]; % MHz
PL = 32.44 + 20*log10(f) + 20*log10(d);
```

```
figure
subplot(2,1,1)
plot(f,PL,'-o','LineWidth',2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
subplot(2,1,2)
bar(PL)
title('Comparison of Propagation Loss Across Bands')
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
xticklabels({'Sub-1GHz','Mid-band','mmWave'})
grid on
```

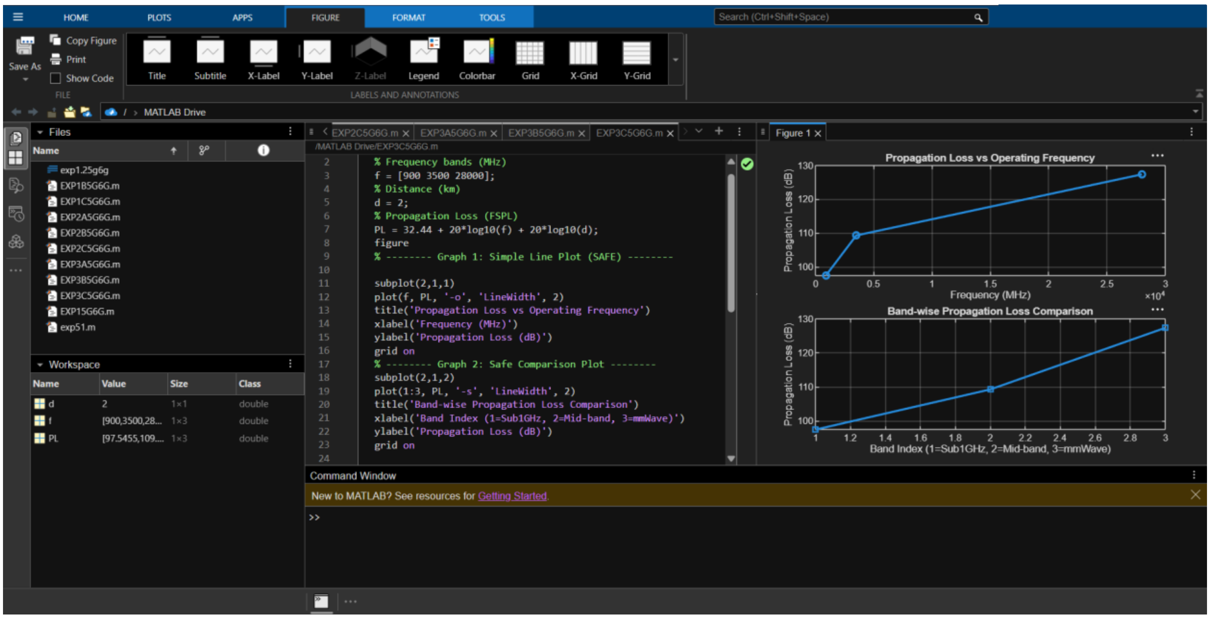
OUTPUT:



Objective 3: Matlab Code and Output

```
clc; clear; close all;

% Frequency bands (MHz)

f = [900 3500 28000];

% Distance (km)

d = 2;

% Propagation Loss (FSPL)

PL = 32.44 + 20*log10(f) + 20*log10(d);

figure

% -------- Graph 1: Simple Line Plot (SAFE) --------


subplot(2,1,1)

plot(f, PL, '-o', 'LineWidth', 2)

title('Propagation Loss vs Operating Frequency')

xlabel('Frequency (MHz)')

ylabel('Propagation Loss (dB)')

grid on

% -------- Graph 2: Safe Comparison Plot --------

subplot(2,1,2)

plot(1:3, PL, '-s', 'LineWidth', 2)

title('Band-wise Propagation Loss Comparison')

xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')

ylabel('Propagation Loss (dB)')

grid on
```

OUTPUT:



Propagation Loss vs Operating Frequency
Frequency (MHz)
Propagation Loss (dB)
Band-wise Propagation Loss Comparison
Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)
% Frequency bands (MHz)
f = [900 3500 28000];
% Distance (km)
d = 2;
% Propagation Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
% -------- Graph 1: Simple Line Plot (SAFE) --------
subplot(2,1,1)
plot(f, PL, '-o', 'LineWidth', 2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% -------- Graph 2: Safe Comparison Plot --------
subplot(2,1,2)
plot(1:3, PL, '-s', 'LineWidth', 2)
title('Band-wise Propagation Loss Comparison')
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
grid on
Command Window